SENTINEL

AI/NLP Risk Engine

From headlines to risk signals: sentiment-driven index rebalancing and event-driven stress testing

Goureesankar S Nair · Vellore Institute of Technology (VIT)

RiskSignal

“Moody's downgrades several mid-size banks after Harbor Point default; bank stocks plunge”

entity	**JPM**
sentiment	**−0.57**
event	**Credit Event (0.99)**
impact	**10.0 / 10**
regions	**North America**
action	**Stress test → CET1 2.5%**

S&P Global & Crisil Campus Hackathon · Phase III Case Study

Markets move on words before numbers, and risk teams need those words as data

The problem

- Thousands of headlines and posts a day; the few that matter (a strait closed, a lender in default, an inflation surprise) are buried in noise
- Scheduled stress tests and manual rebalancing react days after the market already has
- Black-box scores cannot be challenged, which makes them hard to govern in a bank

Our approach



Ingest

News (NewsAPI, GDELT, RSS) + social (Reddit, X-style) through fault-tolerant connectors



Understand

Entity linking → entity-targeted sentiment + event classification



Score

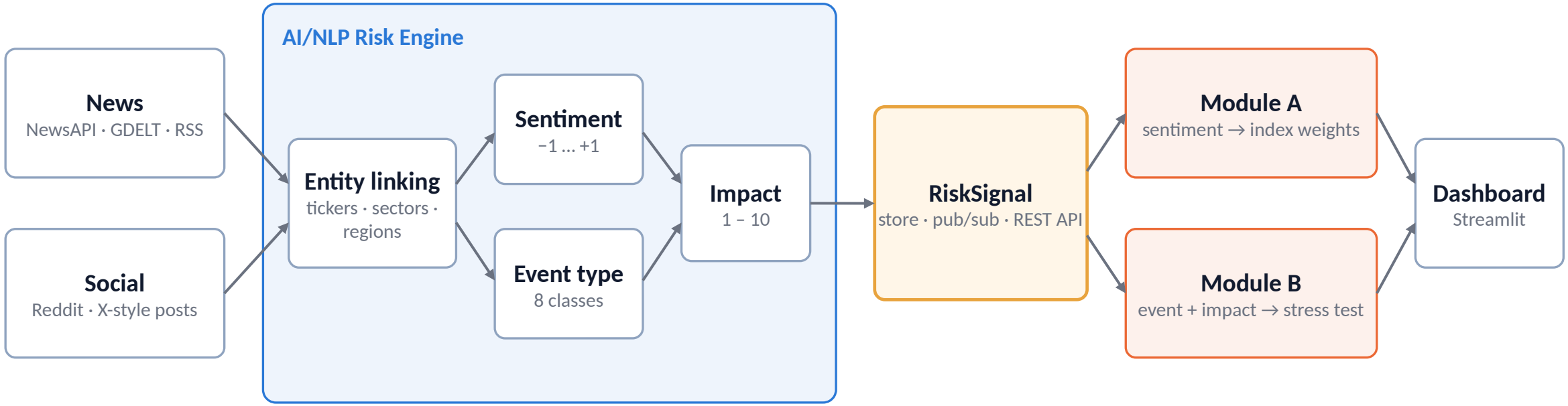
Explainable Impact Score 1–10 with its driver decomposition



Act

Module A tilts the index; Module B stress-tests on impact > 7

One engine, one signal contract, two consumers



Two source types

News and social connectors normalise everything into one Document; live APIs fail soft, and CSV replay keeps the demo deterministic.

Explainable signal

RiskSignal = entity, sentiment, event, impact, sectors, regions, credibility, plus impact drivers and keyword evidence.

Three delivery paths

Pub/sub topic for in-process subscribers, SQLite/JSONL for files, FastAPI REST for any downstream app.

Implementation highlights: tech choices and why

Entity-targeted sentiment

FinBERT when available; offline: Loughran-McDonald + 80+ finance phrase patterns (“inflation surges” is negative) + VADER + TF-IDF/LR. Each ticker is scored on its own sentences.

Event taxonomy

8 classes from weighted regexes in YAML (title counts double) for speed and auditability; a TF-IDF/LR model takes over when the rules are unsure.

Impact score 1–10

Base severity + tone + intensity + confidence + reach + burst, × source credibility. Every term is stored with the signal.

Module A rule

target = base·exp(2·S); S uses a 48h decay and credibility weights; bounds [0.4×, 2×] and ≤18%; 60% smoothing caps turnover.

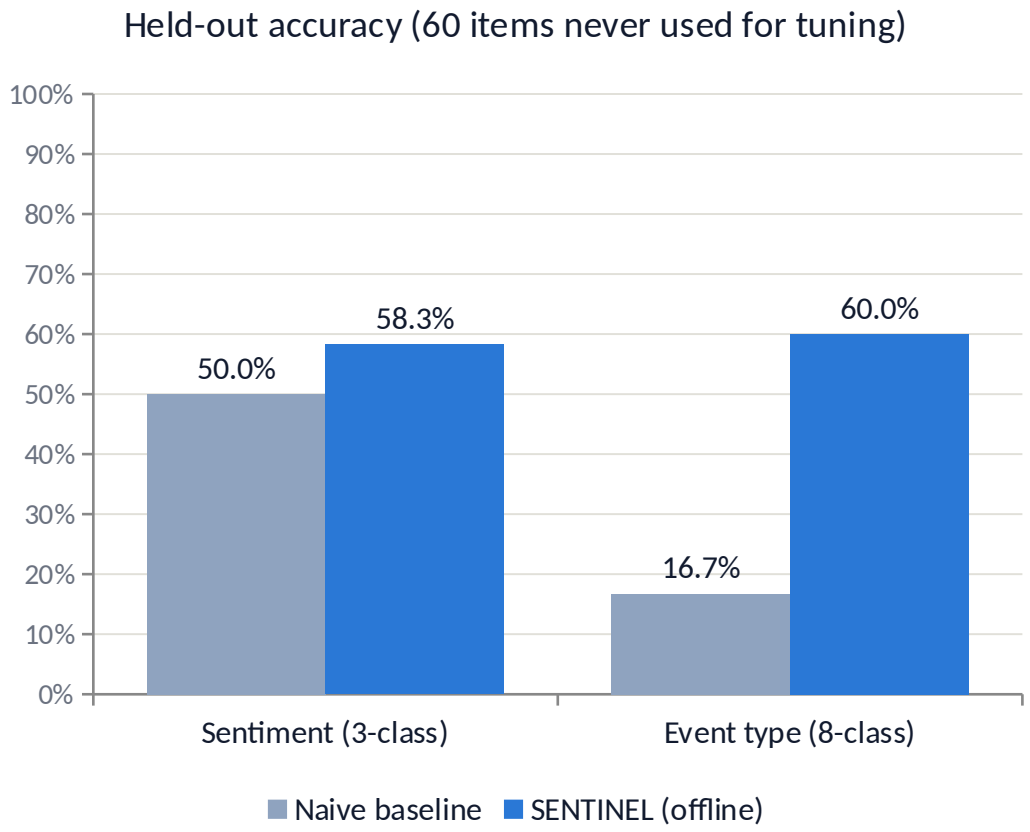
Module B valuation

Bonds: duration + convexity. Loans: ΔPD·LGD and jump-to-default. IRS/CDS: DV01. Plus FX and equity beta. Reports CET1 = capital / RWA.

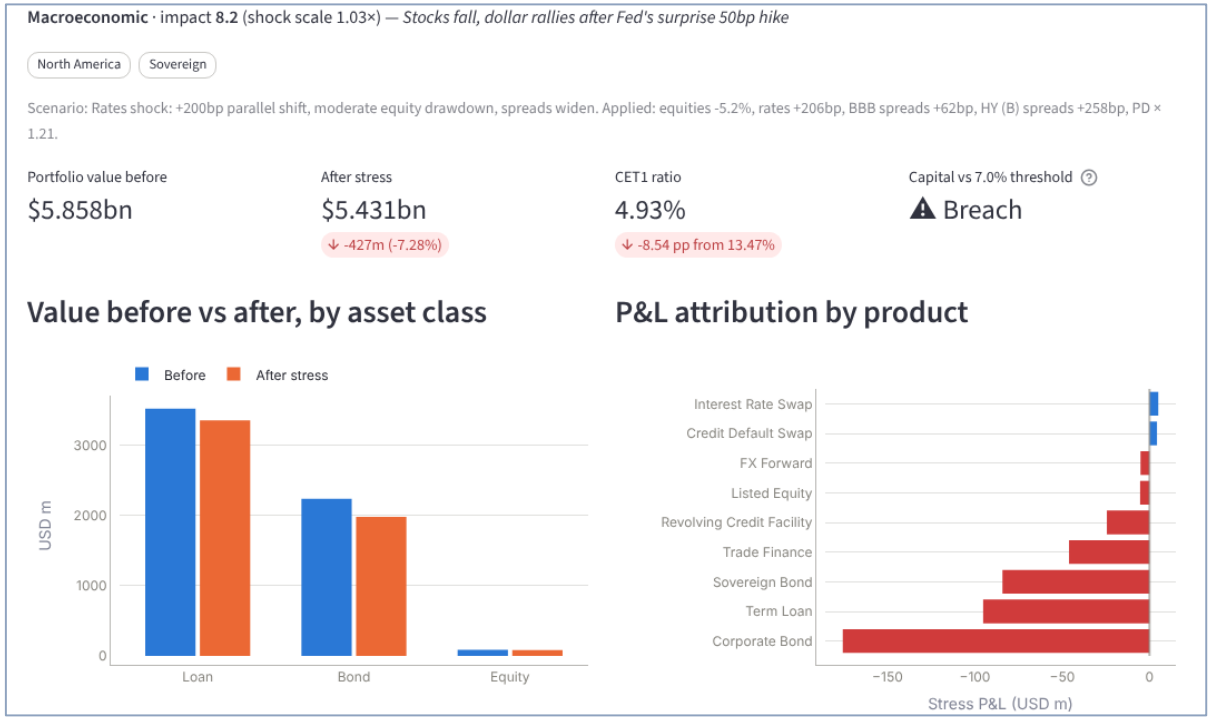
Built to demo & run live

Replay clock streams data in time order with no look-ahead; live mode polls every 5 min; 21 unit tests; one CLI for every task.

Key results: better than naive baselines, and honest about the gap



Baselines: VADER for sentiment, majority class for events. The dev-set score is 92–98%; the gap is why FinBERT is the primary backend.



641 signals

from 625 docs, ~90 docs/s

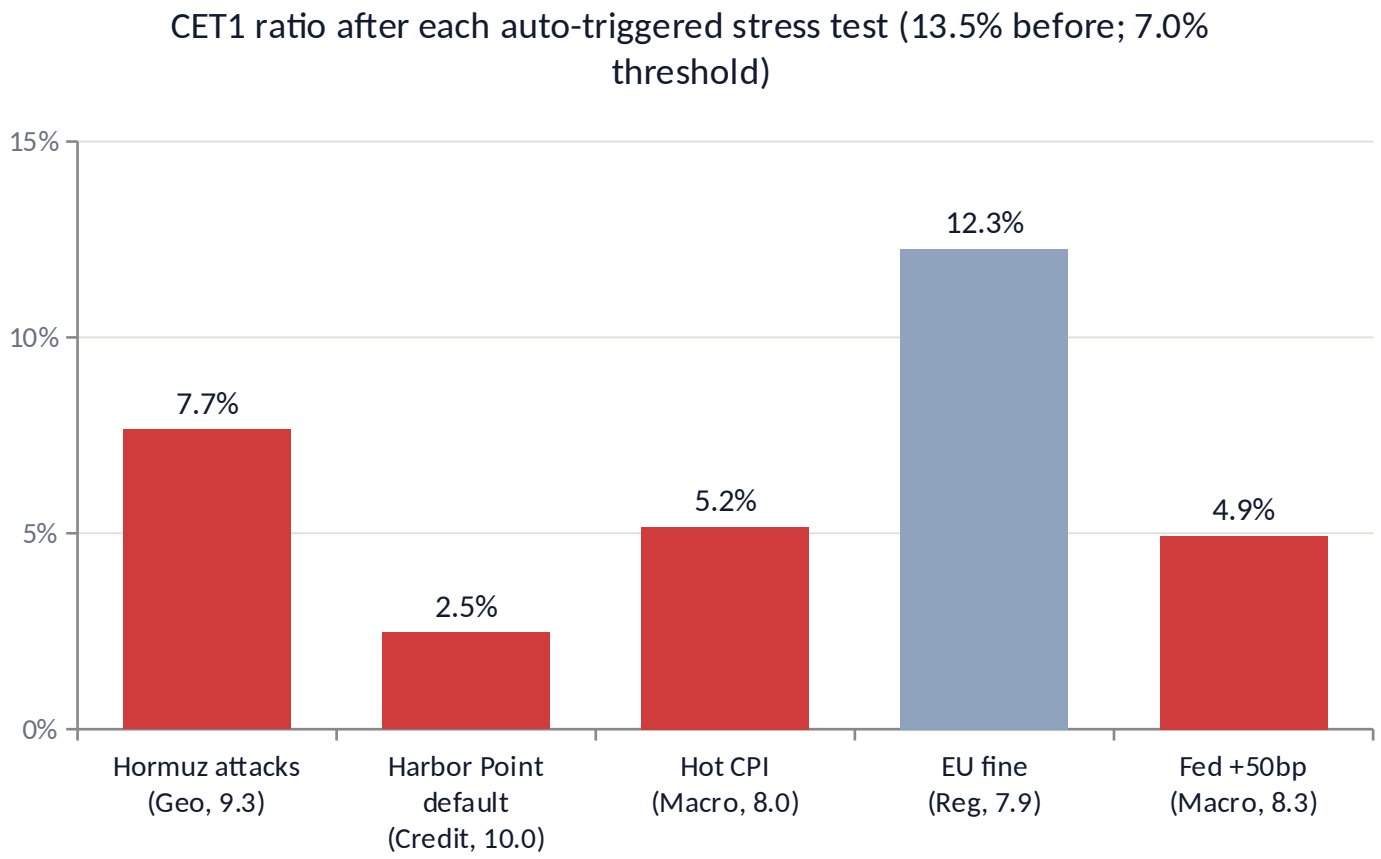
5 stress tests

auto-triggered; ceasefire ignored

+0.29 pts

Module A vs −0.37 naive

Minutes after a headline breaks, the bank knows its capital at risk



Red = below the 7.0% threshold (4.5% minimum + 2.5% buffer). Synthetic USD 5.9bn book, illustrative capital.

Risk manager

An early-warning stress test fires automatically, with P&L down to the position. It surfaced a ~5% single-name concentration and unhedged duration.

Portfolio manager

A systematic, bounded tilt with controlled turnover. It cut JPM from 7.0% to 4.4% at the close on the day of the default headline.

Model governance

Every score has a decomposition and its keyword evidence, so the model can be challenged and audited.

Limitations and next steps

Assumptions & gaps

- Demo stream, prices and banking book are synthetic; Module A returns have an edge by construction
- Offline NLP generalises modestly (58% / 60% held-out); eval sets are small and author-labelled
- Stress model is first-order: no full revaluation, no correlations, linear impact scaling
- Market-wide news does not yet propagate to sector weights in Module A

Next steps

- Fine-tune FinBERT on Financial PhraseBank + labelled social data; benchmark with the evaluate command
- Deduplicate stories across outlets with sentence embeddings to sharpen burst detection
- Backtest Module A on months of live GDELT/NewsAPI signals with real yfinance prices
- Full revaluation, correlated shocks and Kafka streaming for production scale